PROPOSED FINAL-YEAR BE CSE PROJECT

Fuzzy-Causal Evidence Attribution for Explainable RAG-Based Fact-Checking

Research-inspired extension of "Explaining the Role of Evidence in Data-Driven Fact-Checking"

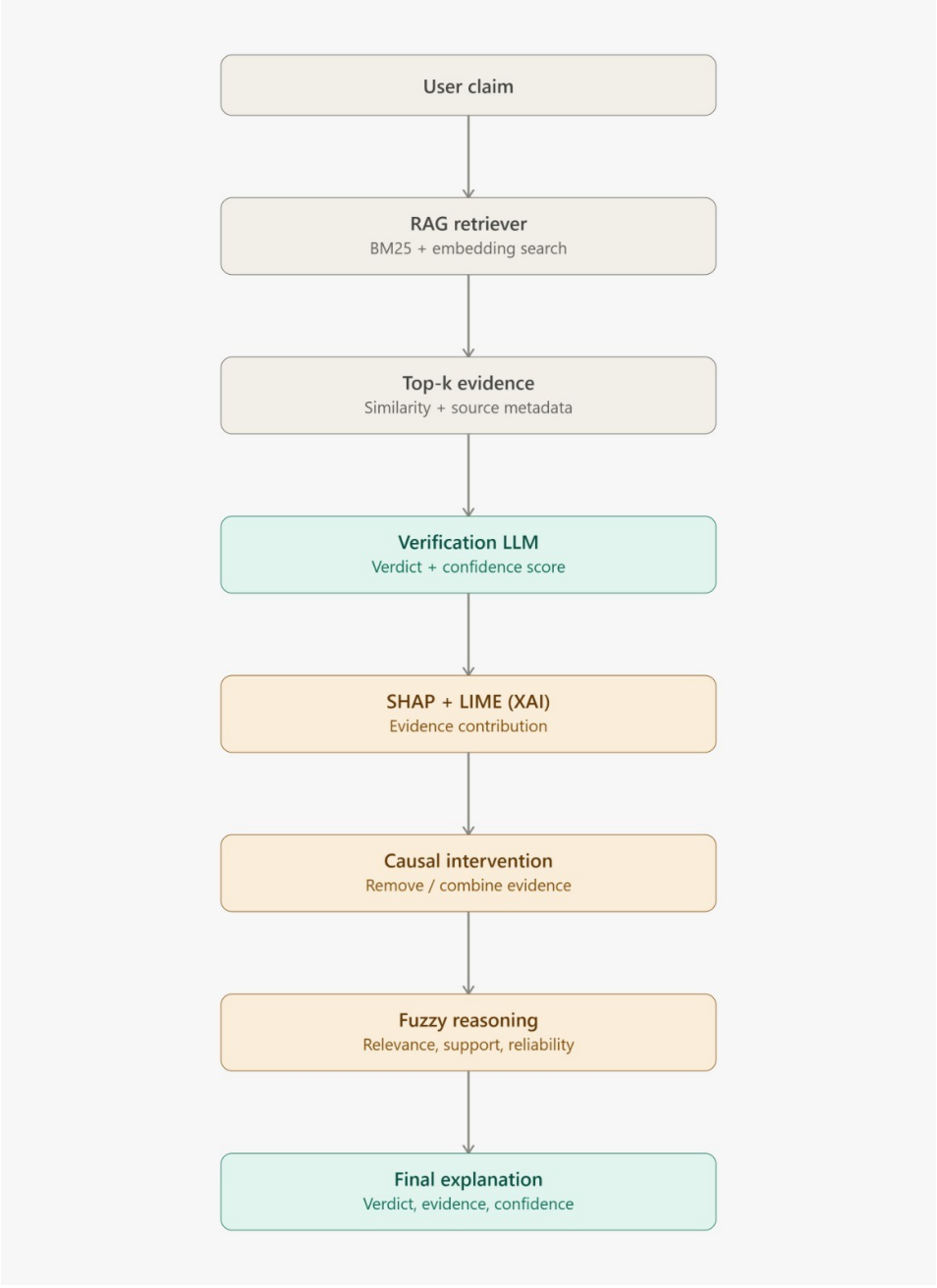


Figure 1. Proposed end-to-end project architecture.

This document describes the proposed solution, its research motivation, system modules, and intended novelty.

1. Project Overview

Core idea. The proposed system is an explainable RAG-based fact-checking platform. Given a natural-language claim, it retrieves evidence, verifies the claim, estimates which evidence influenced the decision, tests whether that evidence actually changes the decision, and produces a reliability-aware explanation rather than only a yes/no verdict.

Research philosophy. The project is designed as an extension of the IEEE paper rather than a direct reproduction. The base paper studies evidence-level attribution with post-hoc methods such as SHAP and LIME. Our project keeps that foundation and adds RAG retrieval, intervention-based influence testing, fuzzy evidence reasoning, conflict handling, explanation stability, and reliability analysis.

2. What the Base IEEE Paper Does

- Models the fact-checking task as claim + evidence -> Supports / Refutes / Not Enough Information.
- Treats individual sentences or table cells as evidence pieces whose contribution can be analyzed.
- Uses LIME and SHAP by sampling different subsets of evidence and observing how the verifier prediction changes.
- Evaluates multiple datasets and verifier models to study evidence attribution and compare post-hoc explanations with native or prompted explanations.
- Shows that post-hoc explanations can be useful, but also highlights limitations around faithfulness, perturbation sensitivity, domain shift, mixed evidence, and reliability analysis.

Important boundary: the original paper is primarily an evaluation/benchmark study of evidence attribution. The proposed project extends that work into a complete retrieval + verification + explanation system.

3. Proposed End-to-End Solution

3.1 User Claim

The user submits a textual claim, for example: “The James Webb Space Telescope was launched in 2021.” The claim is the anchor for retrieval and verification.

3.2 RAG Retriever

A hybrid retrieval layer searches an evidence corpus using BM25 for lexical matching and embedding-based semantic search for meaning-level matching. The two retrieval signals can be fused or used for reranking.

3.3 Top-k Evidence

The system selects the most relevant evidence passages and attaches retrieval metadata such as similarity, rank, source, and source-type information. Evidence is kept as separate pieces so later attribution can be performed at the evidence level.

3.4 Verification LLM

The claim and retrieved evidence are passed to a verification model that returns Supports, Refutes, or Not Enough Information, together with an initial confidence estimate. The verifier is grounded in retrieved evidence instead of answering from free-form model knowledge alone.

3.5 SHAP + LIME XAI

Post-hoc attribution estimates the contribution of each evidence piece. By perturbing evidence subsets and observing prediction changes, the system obtains SHAP/LIME contribution signals that answer: “Which evidence appears to influence the model?”

3.6 Causal Intervention

The project adds an intervention stage. Evidence pieces are removed or combined and the verifier is queried again. The resulting prediction or probability change becomes an intervention-based influence signal. This tests whether an apparently important evidence item actually changes the decision.

3.7 Fuzzy Reasoning

A fuzzy inference layer combines multiple imperfect signals rather than forcing a hard Useful/Noise decision. Inputs can include semantic relevance, SHAP contribution, LIME contribution, causal influence, contradiction, source reliability, and explanation stability. The output can be continuous evidence relevance, support degree, contradiction degree, or explanation reliability.

3.8 Conflict Detection

If evidence pieces support and refute the same claim, the system explicitly models conflicting evidence instead of forcing a binary conclusion. It can separate supporting evidence, refuting evidence, and neutral/contextual evidence.

3.9 Explanation Stability

The system can repeat attribution under controlled perturbation variations and measure how consistent the evidence rankings or contribution scores are. This becomes a stability score for the explanation itself.

3.10 Final Explanation

The final response combines the verdict, confidence, most influential evidence, supporting/refuting evidence, fuzzy evidence scores, causal influence, conflict level, and explanation reliability/stability into an auditable explanation.

4. How Fuzzy Logic Fits the Research

Fuzzy logic is not being added as a decorative component. It is used because evidence quality and model influence are naturally gradual rather than binary.

Example evidence-level signals:

- Relevance = 0.91
- Support = 0.84
- Contradiction = 0.08
- Causal influence = 0.88
- Source reliability = 0.76
- Attribution stability = 0.90

Example fuzzy rules:

- IF relevance is HIGH AND causal influence is HIGH AND source reliability is HIGH, THEN evidence reliability is VERY HIGH.
- IF relevance is HIGH AND contradiction is HIGH, THEN evidence state is CONFLICTING.
- IF SHAP influence is HIGH BUT causal influence is LOW, THEN attribution confidence is LOW.
- IF support is HIGH AND refutation is LOW AND stability is HIGH, THEN evidence confidence is HIGH.

5. Research Gaps Addressed by the Proposed Project

Gap in base paper	Planned project response
SHAP/LIME may reflect correlations or perturbation artifacts instead of truly causal influence.	Add evidence intervention and compare attribution signals with actual decision changes.
Evidence usefulness is often reduced to a binary Useful/Noise view.	Use fuzzy degrees for relevance, support, contradiction, and reliability.
Mixed or conflicting evidence is not fully represented in the benchmark setting.	Add explicit conflict detection and evidence-role analysis.
Attribution stability under domain shift and different retrieval settings needs more study.	Evaluate the system across different domains and retrieval configurations.
Explanation reliability and uncertainty need stronger analysis.	Compute explanation stability, evidence reliability, and confidence-aware outputs.
A verifier can appear to use evidence while also relying on model priors or baseline behavior.	Compare evidence-based attribution with intervention results and expose weakly grounded decisions.

6. Key Research Contributions We Aim to Make

Contribution	What it adds
Hybrid RAG fact-checking	A complete retrieval-to-verification pipeline using lexical + semantic evidence retrieval.
Evidence intervention analysis	A practical causal-style test of evidence influence through removal and combination experiments.
Fuzzy evidence reasoning	Continuous evidence relevance and reliability instead of rigid evidence labels.
Conflict-aware verification	Explicit representation of supporting, refuting, neutral, and conflicting evidence.
Explanation stability	A measurable estimate of how robust an explanation is to perturbation changes.
Reliability-aware final explanation	A user-facing explanation that reports not just a verdict, but the strength and reliability of the evidence behind it.

7. Example End-to-End Output

Claim: “JWST was launched in 2021.”

Verdict: SUPPORTED

Confidence: 94%

Evidence	Role	Relevance	Causal influence
E1: "JWST launched on December 25, 2021."	Supporting	0.96	0.91
E2: "JWST is an infrared space telescope."	Contextual	0.38	0.09
E3: unrelated information	Noise	0.04	0.01

- Most influential evidence: E1
- Explanation reliability: 0.91
- Explanation stability: 0.89
- Conflict level: Low
- Reason: removing E1 changes the verifier from SUPPORTS to NOT ENOUGH INFORMATION, while the other evidence pieces have little effect.

8. Comparison: Base Paper vs. Our Proposed Project

Component	Base paper	Proposed project
Fact verification	Yes	Yes
Evidence-level attribution	Yes	Yes
SHAP	Yes	Yes
LIME	Yes	Yes
RAG retrieval	Not the main contribution	Yes
Hybrid BM25 + embeddings	No	Yes
Causal/intervention testing	No	Yes
Fuzzy evidence reasoning	No	Yes
Continuous evidence relevance	Limited/binary evaluation	Yes
Conflict detection	Limited	Yes
Explanation stability	Identified as concern	Yes
Reliability / uncertainty score	Future direction	Yes
Domain-shift evaluation	Future direction	Yes

9. What Makes Our Project Unique?

The novelty is not any single library or technique. SHAP is not new, LIME is not new, RAG is not new, and fuzzy logic by itself is not a sufficient research contribution. The intended novelty is the way these components are combined around a specific research question: can an evidence-attribution pipeline become more reliable by combining apparent attribution, intervention-based influence, and fuzzy uncertainty?

Base paper asks: "Which evidence appears to influence the model?"

Our project asks: "Which evidence appears influential, does that evidence actually change the decision, and how confident should we be in the resulting explanation?"

- RAG retrieves evidence dynamically rather than assuming a fixed evidence set.
- SHAP/LIME provide attribution signals, while intervention tests provide an additional influence signal.
- Fuzzy reasoning turns multiple imperfect signals into continuous evidence and reliability scores.
- Conflict detection prevents the system from hiding disagreement between sources.
- Explanation stability measures whether the explanation survives controlled perturbation changes.
- Domain-shift and sensitivity experiments provide a research evaluation beyond a single benchmark.

10. Suggested Final System Narrative

One-sentence project description: “An explainable RAG-based fact-checking system that not only determines whether a claim is supported, refuted, or uncertain, but also identifies, tests, ranks, and quantifies the reliability of the evidence responsible for that decision.”

Intended final-year scope. The project is deliberately modular. A functional baseline can be built with pretrained NLP/LLM components, while the research contribution is concentrated in evidence retrieval, attribution, intervention analysis, fuzzy reasoning, conflict handling, and reliability evaluation. This creates a large-looking final-year system without requiring the training of a large model from scratch.